

IB MATH AA SL

Lesson 20

Solving Trigonometric Equations

Trigonometry · 60-minute lesson

Syllabus SL 3.8 · Property of Lynda Badmus Education

θ

Do Now — recall

ON YOUR WHITEBOARD

1. Solve $\sin \theta = 1$ for $0 \leq \theta \leq 360^\circ$.
2. $\sin^{-1}(0.5) = ?$
3. How many solutions can $\sin \theta = 0.5$ have in $0-360^\circ$?

Answers (click to reveal)

1. $\theta = 90^\circ$
2. 30°
3. two

What we're learning today

LEARNING OBJECTIVES

- 1 Solve**
 $\sin \theta = k$, $\cos \theta = k$, $\tan \theta = k$.
- 2 Find**
all solutions in a given interval.
- 3 Use**
the unit circle / CAST for extra solutions.
- 4 Solve**
equations needing rearrangement first.

SUCCESS CRITERIA — I CAN...

- ✓
find the first solution with the inverse function
- ✓
use symmetry to find the second solution
- ✓
list all solutions in the stated interval
- ✓
rearrange before solving where needed

The method

STEPS

$$\theta = \sin^{-1}k, \cos^{-1}k, \tan^{-1}k$$

1. find the first (principal) solution
2. use symmetry / CAST for others
3. keep only those inside the interval

SECOND SOLUTIONS

sin

θ and $180^\circ - \theta$

cos

θ and $360^\circ - \theta$

tan

θ and $\theta + 180^\circ$

Solve over an interval

QUESTION

Solve $\sin \theta = 0.5$ for $0^\circ \leq \theta \leq 360^\circ$.

SOLUTION

Step 1 first solution: $\sin^{-1}(0.5) = 30^\circ$

Step 2 sine: second is $180^\circ - 30^\circ = 150^\circ$

$$\theta = 30^\circ, 150^\circ$$

Both lie in 0° – 360° , so these are the answers.

Your turn — solve

ANSWER IN YOUR BOOK

1. $\cos \theta = 0.5$, 0° – 360° .

2. $\tan \theta = 1$, 0° – 360° .

3. $\sin \theta = 0$, 0° – 360° .

Answers (click to reveal)

1. 60° , 300°

2. 45° , 225°

3. 0° , 180° , 360°

Rearranging first

GET THE RATIO ALONE

$$2 \sin \theta - 1 = 0 \Rightarrow \sin \theta = \frac{1}{2}$$

Isolate sin/cos/tan before using the inverse

Watch

factor if there is a squared term

TIP

Treat $\sin \theta$ like a single unknown.

Quadratic in $\sin \theta$

factor, then solve each piece.

GDC

Paper 2: graph and read intersections.

Rearrange then solve

QUESTION

Solve $2 \cos \theta - 1 = 0$ for $0^\circ \leq \theta \leq 360^\circ$.

SOLUTION

Step 1 isolate $\cos \theta$:

$$\cos \theta = \frac{1}{2}$$

Step 2 first solution $\cos^{-1}(0.5) = 60^\circ$

Step 3 cosine: second is $360^\circ - 60^\circ = 300^\circ$

$$\theta = 60^\circ, 300^\circ$$

Exam-style question

Solve $2 \sin \theta = \sqrt{3}$ for $0^\circ \leq \theta \leq 360^\circ$. [4]

MARK SCHEME — reveal after students attempt (tutor only)

$\sin \theta = \sqrt{3}/2$ (M1) $\rightarrow \theta = 60^\circ$ (A1); second solution $180^\circ - 60^\circ = 120^\circ$ (M1) $\rightarrow \theta = 60^\circ, 120^\circ$ (A1)

Common mistakes



One solution only

Most intervals give 2+ solutions — use symmetry.



Interval

Discard any solution outside the stated range.



Isolate first

Rearrange to $\sin/\cos/\tan = k$ before using the inverse.



Degrees vs radians

Match the interval's units on your GDC.

Mixed practice

QUESTIONS

1. Solve $\sin \theta = \sqrt{2}/2$, 0° – 360° .
2. Solve $\tan \theta = \sqrt{3}$, 0° – 360° .
3. Solve $2\cos \theta + 1 = 0$, 0° – 360° .
4. Solve $\cos \theta = 1$, 0° – 360° .

Answers (click to reveal)

1. $45^\circ, 135^\circ$
2. $60^\circ, 240^\circ$
3. $\cos \theta = -\frac{1}{2} \rightarrow 120^\circ, 240^\circ$
4. $0^\circ, 360^\circ$

Plenary & exit ticket

KEY FORMULAS

$$\sin : \theta, 180^\circ - \theta$$

$$\cos : \theta, 360^\circ - \theta$$

EXIT TICKET — before you leave

1. Solve $\sin \theta = 1$, 0° – 360° .

2. Solve $\cos \theta = 0$, 0° – 360° .

3. Solve $\tan \theta = 0$, 0° – 360° .

Answers (click to reveal)

1. 90°

2. $90^\circ, 270^\circ$

3. $0^\circ, 180^\circ, 360^\circ$

Self-assess:
• need more help
• nearly there
• confident

Homework

SET ON DR FROST MATHS (auto-marked)

Task:

Solving trig equations.

- Aim for 80%+ before the next lesson
- Re-attempt any flagged questions
- Bring one tricky question to discuss

NEXT LESSON

Lesson 21

The sine and cosine rules for non-right-angled triangles.

$$a^2 = b^2 + c^2 - 2bc \cos A$$